

Convex Lens - HKDSE Physics Summary

1. Basic Properties of Convex Lens

1.1 Definition and Symbol

- **Convex Lens = Converging Lens**
- Incident rays parallel to the principal axis **converge** at the focus after refraction
- Symbol: Lens shape with outward curves on both ends (thicker in the middle)

1.2 Key Terms

- **Principal axis** (主軸) : Horizontal line passing through the center of the lens
 - **Optical center C** (光心) : Center point of the lens
 - **Focus F** (焦點) : Point where parallel rays converge after refraction
 - **Focal length f** (焦距) : Distance from optical center to focus
 - **2F**: Point at twice the focal length from the lens
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2. Three Principal Rays for Convex Lens

Rule 1: Parallel in, Focus out

- A ray parallel to the principal axis passes through the focus F on the opposite side after refraction
- Mnemonic: **Parallel in, Focus out**

Rule 2: Focus in, Parallel out

- A ray passing through the focus F emerges parallel to the principal axis after refraction
- Mnemonic: **Focus in, Parallel out**

Rule 3: Straight through center

- A ray passing through the optical center C travels straight through without changing direction
 - Mnemonic: **C in C out, Straight through**
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3. Six Image Formation Cases for Convex Lens

Case 1: Object beyond $2F$

- **Image position:** Between F and $2F$ on opposite side
- **Image nature:** Real, inverted, diminished
- **Application:** Camera, human eye

Case 2: Object at $2F$

- **Image position:** At $2F$ on opposite side
- **Image nature:** Real, inverted, same size
- **Application:** Copying machine

Case 3: Object between F and $2F$

- **Image position:** Beyond $2F$ on opposite side
- **Image nature:** Real, inverted, magnified
- **Application:** Projector, slide projector

Case 4: Object at F

- **Image position:** At infinity
- **Image nature:** No image formed (parallel rays after lens)
- **Application:** Searchlight, flashlight

Case 5: Object inside F

- **Image position:** Same side as object
- **Image nature:** Virtual, erect, magnified
- **Application:** Magnifying glass, reading glass

Case 6: Object at infinity

- **Image position:** At focus F
 - **Image nature:** Real, inverted, highly diminished (point)
 - **Application:** Telescope objective lens
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4. Summary Table of Image Formation

Object Position	Image Position	Image Nature	Magnification	Type	Application
Beyond 2F	Between F and 2F	Real, inverted	Diminished	Real	Camera
At 2F	At 2F	Real, inverted	Same size	Real	Copier
Between F and 2F	Beyond 2F	Real, inverted	Magnified	Real	Projector
At F	Infinity	No image	-	-	Searchlight
Inside F	Same side	Virtual, erect	Magnified	Virtual	Magnifier
At infinity	At F	Real, inverted	Point	Real	Telescope

5. Lens Formula and Magnification

5.1 Lens Formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Sign Convention:

- f
= focal length (**convex lens: +ve**)
- u
= object distance (always +ve)
- v
= image distance (real image: +ve; virtual image: -ve)

5.2 Linear Magnification

$$M = \frac{v}{u} = \frac{h_i}{h_o}$$

Where:

- M
= magnification
- v
= image distance
- u
= object distance
- h_i
= image height
- h_o
= object height

Interpretation:

- $|M| > 1$
: Magnified
 - $|M| = 1$
: Same size
 - $|M| < 1$
: Diminished
 - $M > 0$
: Erect (virtual image)
 - $M < 0$
: Inverted (real image)
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6. Real Image vs Virtual Image

6.1 Real Image

- **Formation:** Actual convergence of light rays
- **Can be projected:** Yes, can be displayed on screen
- **Position:** Opposite side of lens from object
- **Orientation:** Inverted
- **Examples:** Camera image, projector image

6.2 Virtual Image

- **Formation:** Apparent divergence of light rays (extensions intersect)
 - **Can be projected:** No, cannot be displayed on screen
 - **Position:** Same side of lens as object
 - **Orientation:** Erect
 - **Examples:** Magnifying glass image
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7. Applications of Convex Lens

7.1 Camera

- Object beyond $2F$
- Forms real, inverted, diminished image on film/sensor
- Image between F and $2F$

7.2 Projector

- Object between F and $2F$
- Forms real, inverted, magnified image on screen
- Image beyond $2F$

7.3 Magnifying Glass

- Object inside F
- Forms virtual, erect, magnified image
- Image on same side as object

7.4 Human Eye

- Convex lens (eye lens) focuses light on retina
- Object at various distances
- Forms real, inverted image on retina (brain interprets as erect)

7.5 Telescope

- Objective lens: Object at infinity, image at focus
- Eyepiece lens: Acts as magnifying glass

8. Important Concepts

8.1 Relationship Between Object and Image Position

- **Object moves closer to lens** → Image moves away from lens
- **Object moves away from lens** → Image moves closer to lens
- **Object at 2F** → Image at 2F (symmetry point)

8.2 Power of Lens

$$P = \frac{1}{f}$$

Where:

- P
= power of lens (unit: dioptre, D)
- f
= focal length (unit: meter, m)

Note:

- Shorter focal length → Higher power → Stronger converging ability
- Longer focal length → Lower power → Weaker converging ability

9. Problem-Solving Strategies

9.1 Ray Diagram Method

1. Draw at least two of the three principal rays
2. Locate where rays intersect (real image) or where extensions intersect (virtual image)
3. Measure object and image distances, heights

9.2 Formula Method

1. Identify given values:

u

,

f

, or
 v

2. Apply lens formula:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

3. Pay attention to sign conventions

4. Calculate magnification if needed:

$$M = \frac{v}{u}$$

9.3 Sign Convention Checklist

- Convex lens:

f

is positive

- Object distance

u

: always positive

- Real image

v

: positive (opposite side)

- Virtual image

v

: negative (same side)

10. Common Misconceptions

10.1 Misconception 1

✗ **Wrong:** Virtual image is always smaller than object

✓ **Correct:** Virtual image formed by convex lens (object inside F) is magnified

10.2 Misconception 2

✗ **Wrong:** Convex lens always forms real image

✓ **Correct:** Convex lens forms virtual image when object is inside F

10.3 Misconception 3

✗ **Wrong:** Image at 2F is always magnified

✓ **Correct:** Image at 2F is same size as object

11. HKDSE Exam Key Points

11.1 Must Remember

1. **Three principal rays** and their rules
2. **Six image formation cases** and their characteristics
3. **Lens formula:**

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

4. **Magnification:**

$$M = \frac{v}{u} = \frac{h_i}{h_o}$$

5. **Sign conventions:** Convex lens

f

is +ve, real image

v

is +ve, virtual image

v

is -ve

6. **Applications:** Camera, projector, magnifying glass

11.2 Common Question Types

1. **Ray diagrams:** Draw and locate image
2. **Calculations:** Use lens formula and magnification
3. **Image characteristics:** Determine real/virtual, erect/inverted, magnified/diminished
4. **Applications:** Match lens setup to practical use

11.3 Exam Tips

- Always check sign conventions before calculation
- Draw neat ray diagrams with ruler

- Label all key points: F, 2F, C, object, image
 - State image characteristics clearly: real/virtual, erect/inverted, magnified/diminished/same size
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12. Quick Memory Mnemonics

Three Principal Rays:

Parallel in, Focus out
Focus in, Parallel out
C in C out, Straight through

Six Cases (Object Position):

Beyond 2F → Small real image
At 2F → Same size real image
Between F and 2F → Large real image
At F → No image (infinity)
Inside F → Large virtual image
At infinity → Point at F

Real vs Virtual:

Real image: Can project, inverted, opposite side
Virtual image: Cannot project, erect, same side

Study Suggestions:

1. Practice drawing ray diagrams for all six cases
2. Memorize lens formula and magnification formula
3. Understand sign conventions thoroughly
4. Practice calculations with different scenarios
5. Connect theory to real-world applications (camera, magnifier, etc.)